ARCHIMEDES' ANGLE MEASURE AND THE FUNDAMENTAL THEOREM OF TRIGONOMETRY

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Abstract. We try to understand a distant conversation about a trigonometric principle. Although foundational to calculus, the principle is often treated tacitly in both elementary and advanced accounts. The primordial idea of counting rotations makes one aware of the integers. In the same vein, Archimedes' measure of partial rotations hints at the completeness of the real numbers. By a less well-known implication, these ideas also lead to the complex numbers and to the Fundamental Theorem of Trigonometry. What does it all mean? And is it true?

1. Introduction

1.1. **A café conversation.** The place was noisy and memories are fuzzy. Maybe the details were different. Maybe the conversation never took place at all... No matter, we'll try to recount what we remember and interpret what we recount.

———

S: "What is the Fundamental Theorem of Trigonometry?"

S: "Is it $\sin^2\theta + \cos^2\theta = 1$?"

T: "That's famous and precedes the Fundamental Theorem, but it's a corollary."

S: "How about $e^{i\theta} = \cos\theta + i\sin\theta$?"

T: "You're getting warmer, though the coffee is getting colder. This too is a corollary, less distant in time from the Fundamental Theorem."

A server comes and freshens the interlocutors' coffee.

T: (Pauses actively) "You know, *counting rotations* is a primordial idea. (Pauses again.) It makes us aware of the *integers*. By the same token, Archimedes' measure of partial rotations leads to the *completeness* of the *real numbers*."

T: " . . . eh, . . ., though less well-known, Archimedes' angle measure also unearths the notion of *complex numbers*, paving the way to the Fundamental Theorem of Trigonometry.

(T pauses to sip some coffee.)

T: "Okay, here it is."

Theorem (Fundamental Theorem of Trigonometry)**.** *There is an additive non-constant continuous map $P(\theta)$ of the real line to the unit circle, unique up to rescaling. This map is surjective.*

By *additivity* we mean addition of real numbers corresponds to addition of angles, or equivalently to multiplication on the unit circle (4.1).

1.2. **Stepping back, seeking perspective.** Early in the development of trigonometric calculus we come across the identity

$$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1. \tag{1.1}$$

But what exactly is θ?

A mystifying feature of today's calculus books is their lack of discussion of what exactly is meant by the measure of an angle. The subject is assumed to have been treated somehow in secondary school. However, this is not so, as axiomatic treatments [1] of angle measure bypass this issue by design.

The notion of *solid angle* may get an entire chapter in an advanced monograph [2]. Yet in pre-calculus or early portions of calculus treatments, instead of developing angle measure within the framework of just-learned Cartesian geometry, books often go outside this framework by appealing to pictures not only for motivation, but also for justification. As a consequence, one forfeits the opportunity to present angle measure as a basic paradigm of calculus.

Typically, calculus books state that the measure θ of an angle is the length of the subtended arc along the unit circle, use this to define $\sin\theta$, then go on to derive (1.1) by sandwiching a sector's area between the areas of inscribed and circumscribed triangles. At the beginning of Cartesian geometry, none of this has yet been defined analytically [11].

The book [13] comes close to defining the measure of an angle, as it discusses the history of the subject at length. The books [8], [9] ignore the issue completely, although the material here would fit perfectly there.

Whatever definition one takes for the measure of the angle, it should be *additive*: When angles are stacked, their measures should add (Figure 1).

Additive angle measure θ was used implicitly by Archimedes (287 – 212 BCE), as part of his work [7] leading to his estimate of the half-circumference of the unit circle,

$$\frac{223}{71} < \pi < \frac{22}{7}. \tag{1.2}$$

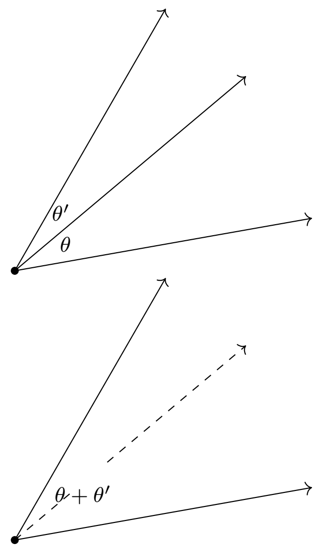

FIGURE 1. Additivity.

By contrast, Hipparchus (190 – 120 BCE) [14] and Ptolemy (100 – 170 CE) [15] used chord measure θ_1, which is equivalent to θ but not additive, to build their trigonometric tables. Aryabhata (476 – 550 CE) presented the first additive trigonometric tables [12]. His use of *ardha-jya* for half-chords, shortened to *jya* or *jiva*, morphed into the Arabic *jaib* ("pocket") which translates into the Latin *sinus* [10].

The path below is in three stages. We first introduce angle stacking and show how it leads naturally toward the complex-number multiplication law. We then follow Archimedes in constructing additive angle measure and proving its existence and uniqueness. Finally, we recover trigonometry from angle measure and establish the Fundamental Theorem of Trigonometry.

2. Geometry—angles, and how To stack them

Working in the Cartesian plane, we will take points to be ordered pairs of numbers, say $P = (x, y)$, $P' = (x', y')$, $P'' = (x'', y'')$. Points may be added, $P + P' := (x + x', y + y')$, and scaled by numbers t, $tP := (tx, ty)$.

An *angle* is a pair of rays emanating from a point, the vertex of the angle. If the vertex is the origin $O = (0, 0)$, then an angle is determined by the intersections of its rays with the unit circle. If the points of intersection are denoted N and P, angles are well-defined even when $P = \pm N$. Notice that angles so-defined are *unoriented*—if we swap rays we obtain the same angle. If we choose an ordering for the rays then we obtain the notion of *oriented angle*. We will be somewhat cavalier about the oriented/unoriented dichotomy of angles, except where precision is required.

In what follows, we make the normalizing assumption that $N = (1, 0)$. Readers needn't worry—generality will not be lost. With this normalization, an angle is determined by a single point P on the unit circle, together with ordering. We need to choose an ordering for the elements of $\{N, P\}$, or eschew orientation.

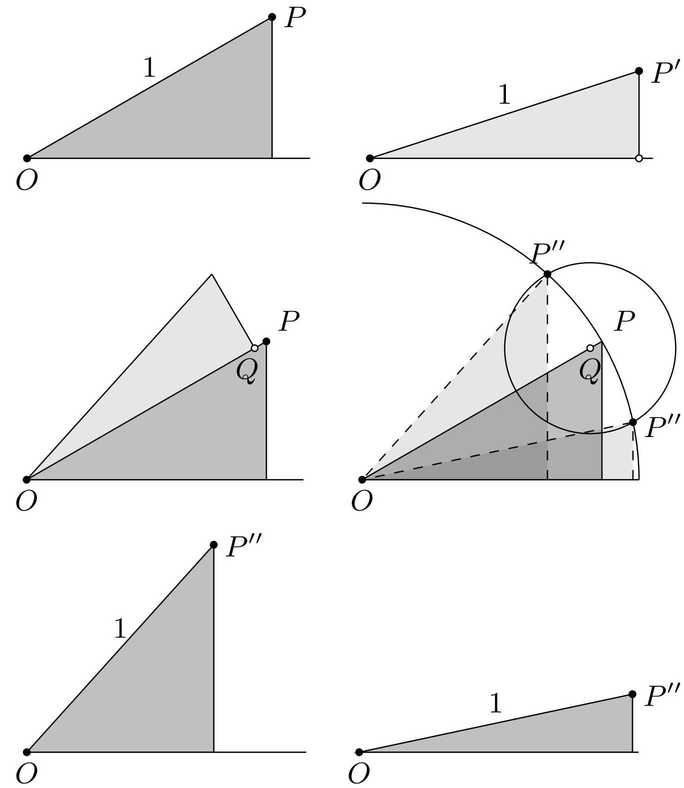

FIGURE 2. Angle stacking.

In the Cartesian plane, angles may be stacked, leading to complex multiplication and division. When stacking two unoriented angles, it's not clear whether we are adding or subtracting them since, for $\angle\alpha$ unoriented, $\angle\alpha = -\angle\alpha$, so $\angle\alpha + \angle\beta$ actually stands for $\pm\angle\alpha \pm \angle\beta$. This is already manifested in ancient tables, such as Ptolemy's [15], where both sums and differences of angles are tabulated.

Let P and P' be points on the unit circle corresponding to two given angles. Then the angles may be stacked (Figure 2), resulting in the angle corresponding to a point P'' on the unit circle. The illustration seems to suggest subliminal orientation, denoting a sum and not a difference. But not to worry, the geometry and the algebra will ally to bring out both possibilities together.

To make this precise, draw the circle with center $Q = x'P$ and radius $|y'|$. Note that in the figure, x' is only slightly smaller than 1, the length of the hypotenuse of triangle P. The small gap is significant, as Q and P play distinct roles.

When $x'y' \neq 0$, we claim this circle intersects the unit circle at two points. For *beneficial ambiguity*[1], we'll label both as P'',

$$\begin{aligned} P'' &= (xx' - yy', x'y + xy') \\ P'' &= (xx' + yy', x'y - xy'). \end{aligned} \tag{2.1}$$

[1]See the definition of *beneficial insects* in the Old Farmer's Almanac. This is only jargon, not meant to denigrate any insects.

Since we obtain two points P'', angle stacking in fact results in two angles, which may remind the reader of the complex product and complex quotient of P and P'. Although we do not develop complex numbers here, we'll speak of the *real* and *imaginary* parts of a point in the plane.

To derive (2.1), let $\langle P, P' \rangle = xx' + yy'$, $|P|^2 = \langle P, P \rangle$, and $P^\perp = (-y, x)$. Then P is on the unit circle if and only if $|P|^2 = 1$, and P, P' on the unit circle satisfy $\langle P, P' \rangle = 0$ if and only if $P' = \pm P^\perp$.

Since P'' is on the circle of center Q and radius $|y'|$, we may write $P'' = Q + y'R$ for some point R on the unit circle. Then

$$\begin{aligned} 1 = |P''|^2 &= |x'P + y'R|^2 \\ &= {x'}^2|P|^2 + {y'}^2|R|^2 + 2x'y'\langle P, R \rangle \\ &= ({x'}^2 + {y'}^2) + 2x'y'\langle P, R \rangle \\ &= 1 + 2x'y'\langle P, R \rangle, \end{aligned}$$

which occurs if and only if $\langle P, R \rangle = 0$, i.e., if and only if $R = \pm P^\perp$. This yields $P'' = x'P \pm y'P^\perp$, which is (2.1). That establishes the claim.

Given any two points $P = (x, y)$, $P' = (x', y')$ on the unit circle, we can define their product $P'' = PP'$ and quotient $P'' = P/P'$ by (2.1). One can check directly that this multiplication rule is commutative and associative, with N as an identity element. (Just copy and paste from an introduction to complex numbers, but don't tell.) Given a third point P'', we proclaim

$$PP' = P'P, \quad P(P'P'') = (PP')P'', \quad PN = NP = P.$$

If $\bar{P} := (x, -y)$, then $P\bar{P} = N$ for any point P on the unit circle, and $P/P' = P\bar{P}'$ is the hermitian product.

In the next section, we define a real-valued additive angle measure $\theta(P)$ for $P \neq -N$. To take into account both cases in (2.1), we require

$$\theta(P) = -\theta(\bar{P}), \quad \text{for} \quad P \neq -N, \tag{2.2}$$

and consequently, $\theta(N) = 0$. We then interpret Figure 1 by

$$\theta(PP') = \theta(P) + \theta(P'), \qquad x > 0, x' > 0. \tag{2.3}$$

It is in this form that we shall establish additivity of $\theta = \theta(P)$.

Note that (2.3) implies

$$\theta(P^2) = 2\theta(P), \qquad x > 0. \tag{2.4}$$

Then (2.3), (2.4) together imply (2.2).

For P on the punctured unit circle $\{P \mid P \neq -N\}$, write

$$\sqrt{P} = \left(\frac{1 + x}{\sqrt{2 + 2x}}, \frac{y}{\sqrt{2 + 2x}} \right). \tag{2.5}$$

Then $P \mapsto \sqrt{P}$ maps the punctured unit circle continuously and bijectively onto the right-half unit circle $x > 0$, with inverse $P \mapsto P^2$, and the imaginary parts of P and $\sqrt{P}$ have the same sign.

Since $(\sqrt{P}\sqrt{P'})^2 = PP'$,

$$\sqrt{P_1 P_2} = \sqrt{P_1}\sqrt{P_2}, \tag{2.6}$$

up to sign, whenever both sides of (2.6) are defined. When the imaginary parts of P, P', PP' have the same sign, the imaginary parts of the sides of (2.6) have the same sign. Hence (2.6) is correct as written, when the imaginary parts of P, P', PP' have the same sign.

3. Measurement

3.1. **Enter Archimedes.** Archimedes interpreted angle measure θ as the length of the arc along the unit circle subtended by the angle (Figure 4). In his treatises *On the sphere and the cylinder* and *Measurement of a circle* [7], Archimedes grappled with the ideas underlying the length, area, and volume of curves, surfaces, and solids.

Archimedes not only computed the length, area, and volume measures of a circle, a sphere, and a ball, he isolated *a priori* properties such measures ought to have. This led to a plethora of ideas including, explicitly or implicitly, the definition of a *convex* curve, the Archimedes Postulate, the additivity of angle measure (§2), the bisection method (§3.2), and the completeness of the real number line.

Archimedes' postulate, often called the *Archimedes convexity postulate* [4], [5], states:

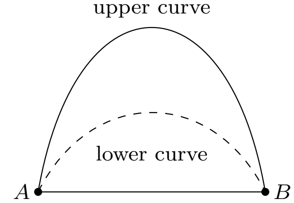

FIGURE 3. Archimedes' convexity postulate.

If two convex curves have common endpoints A, B along a base segment and one curve lies below the other then the lower curve has smaller length than the upper curve.

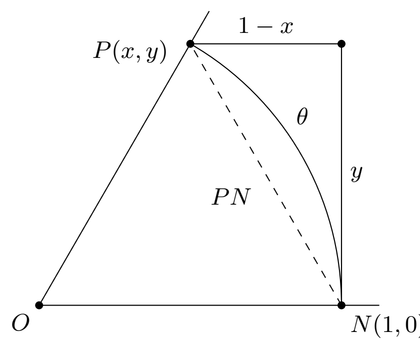

FIGURE 4. Elbow Bounds.

In Figure 4, applying Archimedes' axiom (with base PN) to the curve θ (the lower curve) and the elbow consisting of $1 - x$ and y (the upper curve), we obtain

$$\theta < 1 - x + y\,;$$

applying the axiom to the chord PN and the arc θ we obtain $PN < \theta$. Then, by the Pythagorean theorem, $y < PN$. Putting these together we have

$$y < \theta < 1 - x + y, \quad x, y > 0 \quad \textit{(Elbow Bounds)}. \tag{3.1}$$

The role of Archimedes' postulate ends here: the postulate serves to certify that the Elbow Bounds capture what arc length ought to satisfy. Below we derive the existence and uniqueness of an additive angle measure θ satisfying the Elbow Bounds.

3.2. **Archimedean bisection: chord estimates.** Archimedean bisection is based on Figure 5, a starting point for the *method of exhaustion* of Euclid and Eudoxus. Figures 5, 6, and 7 are in Archimedes' *On the sphere and the cylinder* (see the figures of Propositions 1–6 therein) and his *Measurement of a circle.*

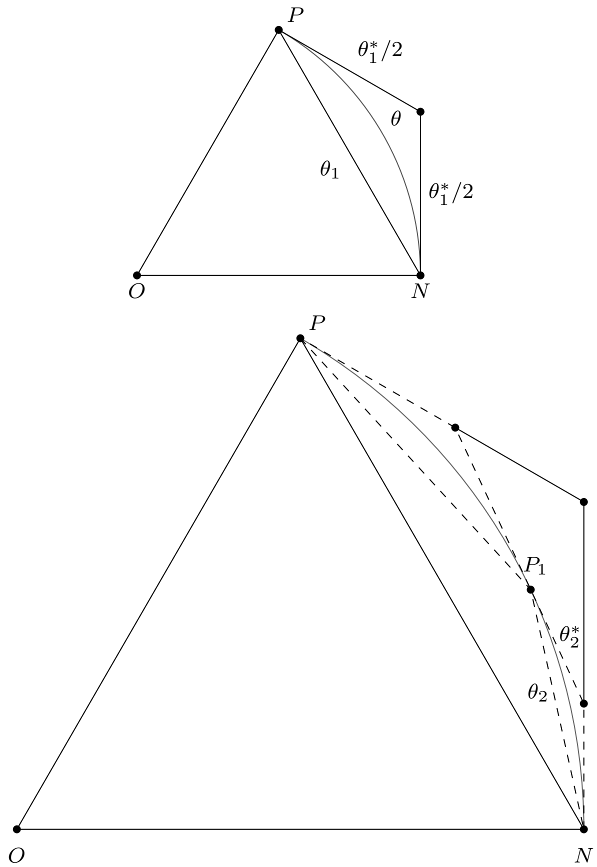

FIGURE 5. Inscribed and circumscribed chords; lower diagram enlarged.

In Figure 5, let N, P_1, P be three points on the unit circle. Let θ_1, θ_1^* be the inscribed and circumscribed chord-length sums on the top, and let θ_1, θ_2, θ_2^*, θ_1^* be the inscribed and circumscribed chord-length sums on the bottom. By the triangle inequality, or by Archimedes' Postulate applied to the four polygonal convex curves joining the endpoints N and P,

$$\theta_1 < \theta_2 < \theta_2^* < \theta_1^*.$$

By incrementally adding points on the arc, partitioning sarcs between adjoining points, we obtain inscribed and circumscribed polygonal approximations of the arc whose lengths are arbitrarily close to each other. This is Proposition 3 in *On the sphere and the cylinder*. By adding points repeatedly to the arc, we have the Archimedes sequences

$$\theta_1 < \theta_2 < \theta_3 < \cdots < \theta_3^* < \theta_2^* < \theta_1^*. \tag{3.2}$$

Now assume the points $P, P_1, P_2, P_3, \ldots$ are generated by bisection (Figure 6): P_1 bisects the arc from N to P, and each P_n bisects the arc from N to P_{n-1}. Thus P_n cuts off $1/2^n$ of the arc from N to P, and the corresponding chords NP_n shrink accordingly. In particular, with $P_n = (x_n, y_n)$, the sequences x_n and y_n are increasing and decreasing respectively.

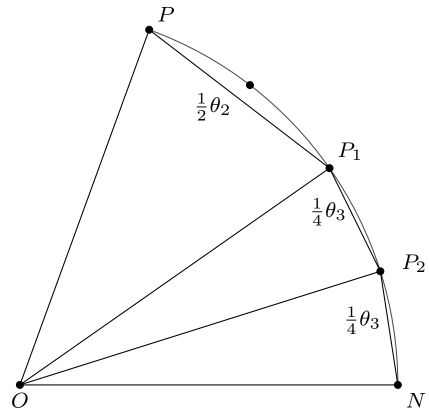

FIGURE 6. Bisection.

We first treat a single bisection. In Figure 7, let P_1 be the midpoint of the arc NP and M_1 the midpoint of the chord PN, and let $P = (x, y)$, $P_1 = (x_1, y_1)$, $Q = (x, 0)$, $Q_1 = (x_1, 0)$. Then P_1 is well-defined when $P \neq -N$.

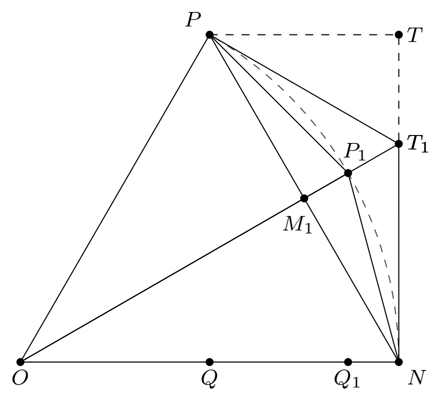

FIGURE 7. First Cut.

Let $\theta_1 = \theta_1(P)$ be the length of PN. Then $\theta_1 > PQ = y$ and $\theta_1 = 2\,M_1N$. Since the triangles OQ_1P_1 and OM_1N are congruent, $M_1N = P_1Q_1 = y_1$, which implies $\theta_1 = 2y_1$. Since $\theta_1^* = PT_1 + T_1N = 2T_1N = 2y_1/x_1$, we conclude $\theta_1^* = \theta_1/x_1$. By Figure 7,

$$\theta_1^* < 1 - x + y = PT_1 + T_1N < PT + TT_1 + T_1N = 1 - x + y,$$

so

$$y < \theta_1 < \theta_1^* < 1 - x + y, \qquad x > 0, y > 0 \tag{3.3}$$

By repeated bisection the arc from N to P is divided into 2^n equal sub-arcs, and, as in the congruence of Figure 7, the corresponding inscribed sub-chords are all equal. Iterating the relations above over these sub-arcs, with $P_n = (x_n, y_n)$ and θ_n, θ_n^* the inscribed and circumscribed chord-length sums, we obtain

$$\theta_n^* = \frac{\theta_n}{x_n} \quad \text{and} \quad \theta_n = 2^n y_n, \qquad n \geq 1. \tag{3.4}$$

The right hand identity in (3.4) connects with Aryabhata's half-chords [12].

3.3. **Uniqueness of angle measure.** *There is at most one real-valued additive angle measure θ on the punctured unit circle satisfying* (3.1).

To establish uniqueness of angle measure, let μ and ν be real-valued additive angle measures satisfying (3.1). By additivity (2.3), it is enough to show $\mu(P) = \nu(P)$ for P in the unit circle first quadrant. Replacing P by P_n in (3.1),

$$2^n y_n < 2^n \mu(P_n) < 2^n(1 - x_n + y_n)\,,$$

and

$$2^n y_n < 2^n \nu(P_n) < 2^n(1 - x_n + y_n).$$

By additivity (2.3), $2^n\mu(P_n) = \mu(P)$ and $2^n\nu(P_n) = \nu(P)$. This implies

$$\begin{aligned}|\mu(P) - \nu(P)| &< 2^n(1 - x_n)\\ &= 2^n \frac{y_n^2}{1 + x_n}\\ &< 2^{-n}\theta_n^2\\ &< 2^{-n}\theta_1^{*2}, \quad \text{for } n \geq 1.\end{aligned}$$

Hence μ and ν are arbitrarily close. This establishes uniqueness of angle measure.

Remark. *Note that we did not require continuity.*

3.4. **Existence of angle measure.** *There is a real-valued additive angle measure θ on the punctured unit circle satisfying* (3.1).

Turning to existence, the sequences in (3.2) are bounded. By (3.4), $y_n \to 0$, hence $x_n \to 1$, as $n \to \infty$. Since the real line is complete, the sequences in (3.2) have a common limit $\theta = \theta(P)$.

Extend $\theta(P)$ to the lower-half unit circle $y < 0$ by $\theta(P) = -\theta(1/P)$, and set $\theta(N) = 0$. Then, by construction, (2.2) and (2.4) are valid. Moreover, by (3.3), θ satisfies (3.1), and $\theta = \theta\left(\sqrt{1-y^2}, y\right)$ is an odd function of y on the right-half unit circle. We now establish additivity of θ.

Let $P'' = PP' = (x'', y'')$, and let $\theta = \theta(P)$, $\theta' = \theta(P')$, and $\theta'' = \theta(P'')$. Let $P_n' = (x_n', y_n')$, $P_n'' = (x_n'', y_n'')$, $n \geq 1$, be the bisection sequences starting from P', P'' respectively, with θ_n', θ_n'', $n \geq 1$, the corresponding chord-length sums.

First assume $yy' > 0$. By (2.2), (2.3) is valid for P, P' if and only if (2.3) is valid for $\bar{P}$, $\bar{P}'$. Hence we may assume P, P' are strictly in the first quadrant. It follows P_n, P_n' are strictly in the first quadrant for $n \geq 1$. By induction and (2.6), $P_n'' = P_n P_n'$. By (2.1), $y_n'' = x_n' y_n + x_n y_n'$, hence, by (3.4),

$$\theta_n'' = x_n'\theta_n + x_n\theta_n', \qquad n \geq 1. \tag{3.5}$$

Since $x_n \to 1$, $x_n' \to 1$, we obtain $\theta'' = \theta + \theta'$, which is (2.3).

Second assume $yy' < 0$. Then $x'' = xx' - yy' > 0$. Since $yy' < 0$, $y''(-y)$ and $y(-y')$ have opposite signs. By switching the roles of P and P' if necessary, we may assume $y''(-y) > 0$. Applying the first case to P'' and $\bar{P}$,

$$\theta(P') = \theta(P''/P) = \theta(P'') + \theta(\bar{P}) = \theta(PP') - \theta(P),$$

which is (2.3). This establishes existence of angle measure.

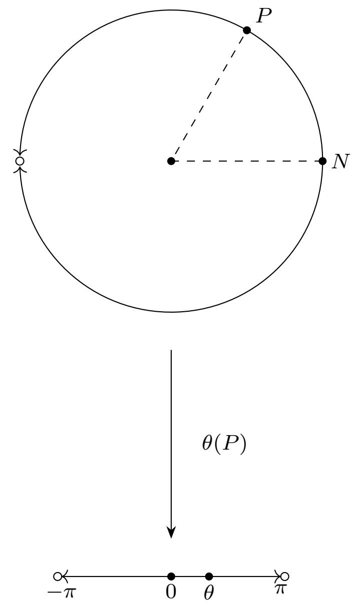

FIGURE 8. Angle measure

3.5. **Properties of the function θ.** *The function $\theta(P)$ is a nondecreasing function of y, for P in the right-half unit circle.*

Let P be in the unit circle first quadrant. From (2.5), y_1 is an increasing function of y. Similarly, with y_2 playing the role of y_1, y_2 is an increasing function of y_1, hence an increasing function of y. Continuing in this manner, y_n, $n \geq 1$, are increasing functions of y. By (3.4), θ_n, $n \geq 1$, are increasing functions of y. Passing to the limit, and since $\theta(P)$ is an odd function of y, the result follows.

3.6. **The number π.** Define $\pi = 2\theta(J)$ where $J = (0, 1)$. Since $2\theta_1(J) = 2\sqrt{2}$ and $x_1(J) = 1/\sqrt{2}$, by (3.2), $2\sqrt{2} < \pi < 4$. To achieve (1.2) using (3.2), Archimedes effectively calculated a variant of

$$2\theta_6(J) = 64\sqrt{2 - \sqrt{\sqrt{\sqrt{\sqrt{\sqrt{2}+2}+2}+2}+2}}.$$

By (3.1), and since $\theta(P)$ is an odd function of y, $\theta(P)$ is continuous at $P = N$ and $\theta(P) \neq 0$ when $P \neq N$. If $P \neq P'$ are in the right-half unit circle and close to each other, then $P/P' \neq N$ and close to N. By (2.3),

$$\theta(P/P') = \theta(P) - \theta(P'), \qquad x_1 > 0, x_2 > 0.$$

It follows that $\theta(P)$ is continuous and injective on the right-half unit circle. By (2.4), $\theta(P)$ is continuous and injective on the punctured unit circle.

Since $\theta(P)$ is an increasing function of y and $\theta(\pm J) = \pm\pi/2$, $\theta(P)$ maps the right-half unit circle into $(-\pi/2, \pi/2)$. By the intermediate value theorem, $\theta(P)$ is a continuous bijection from the right-half unit circle onto $(-\pi/2, \pi/2)$. By (2.4) again, $\theta(P)$ is a continuous bijection from the punctured unit circle $P \neq -N$ onto $(-\pi, \pi)$ (Figure 8).

4. Trigonometry

The basic result of trigonometry, the Theorem in the introduction, is the existence and uniqueness of a continuous map $P(\theta) = (\cos\theta, \sin\theta)$ of the real line into the unit circle satisfying the addition formula

$$P(\theta)P(\theta') = P(\theta+\theta'), \qquad \text{for all } \theta, \theta'. \tag{4.1}$$

The full proof is in the appendix, below we give a sketch.

This form of the addition formula is implicit in the work of de Moivre. Ancient addition formulas are found in the Almagest [15], albeit only for chord lengths.

Separating the addition formula into real and imaginary parts yields the usual trigonometric identities for $\sin\theta$ and $\cos\theta$. By (4.1), any such map satisfies $P(0) = N$ and $P(-\theta) = \bar{P}(\theta)$.

Let $P(\theta)$ be the inverse of angle measure $\theta(P)$. Since $\theta(P)$ maps onto $(-\pi, \pi)$, $P(\theta)$ is defined only on $(-\pi, \pi)$. Since (2.3) is valid only on the right-half unit circle, at this stage we only know that $P(\theta)$ satisfies (4.1) on $(-\pi/2, \pi/2)$.

By restricting to $\theta = \theta'$ in (4.1), we have

$$P(\theta) = P(\theta/2)^2. \tag{4.2}$$

To extend $P(\theta)$ to all reals, we use (4.2) repeatedly, each time doubling the interval of definition of $P(\theta)$. This leads to a map of the real line onto the unit circle satisfying (4.1).

If $P(\theta)$ satisfies (4.1), then $P(\alpha\theta)$ satisfies (4.1), for any real α. Hence, at best, there is uniqueness up to rescaling. To show that this is indeed the case, we need to attach a scale to each such map $P(\theta)$. This scale, or yardstick, is its least positive period.

Given a continuous map $P(\theta)$ satisfying (4.1), we say a real ω is a *period* if $P(\omega) = N$. [2] Then 0 is a period, every integer multiple of a period is a period, and a limit of periods is a period.

Armed with this, uniqueness falls into three parts: First we establish the existence of a positive period, then we establish the existence of a least positive period ω, and use it to rescale the map, then we use (4.1) to show the rescaled map satisfies

$$P(\theta\pi/2) = J^\theta, \qquad J = (0, 1). \tag{4.3}$$

This determines $P(\theta)$ uniquely, completing the sketch.

Inserting $P = (\cos\theta, \sin\theta)$ in the Elbow Bounds (3.1) leads to (1.1) as follows. The left inequality gives $\sin\theta < \theta$, or

$$\frac{\sin\theta}{\theta} < 1.$$

The right inequality gives $\theta < 1 - \cos\theta + \sin\theta$, or

$$\begin{aligned} 0 < 1 - \frac{\sin\theta}{\theta} &< \frac{1-\cos\theta}{\theta} \\ &= \frac{1-\cos\theta}{\theta}\,\frac{1+\cos\theta}{1+\cos\theta} \\ &= \frac{\sin^2\theta}{\theta(1+\cos\theta)} \\ &< \frac{\theta}{\sin\theta}\cdot\frac{\sin^2\theta}{\theta(1+\cos\theta)} \\ &= \frac{\sin\theta}{1+\cos\theta}, \end{aligned}$$

and this last expression tends to zero as $\theta \to 0$. The reasoning here depends on the Elbow Bounds (3.1), hence requires $0 < \theta < \pi/2$ and proves the right hand limit portion of (1.1); that $\sin\theta/\theta$ is an even function verifies (1.1) in full. As a byproduct, we also obtain

$$\lim_{\theta\to 0} \frac{1-\cos\theta}{\theta} = 0.$$

With this limit and (1.1) established, the derivative formulas of trigonometric functions can be computed immediately in the standard way.

Archimedes Appendix

Proof of existence of $P(\theta)$ satisfying (4.1). $\theta(P)$ is a continuous bijection from the punctured circle to $(-\pi, \pi)$, with inverse $P(\theta)$. Restricting $\theta(P)$ to the right half-circle and using the y coordinate as a parameter, $\theta(P)$ is nondecreasing, hence $\theta(P)$ can be viewed as a continuous, monotone function onto $(-\pi/2, \pi/2)$; injectivity shows that the monotonicity is strict. The inverse of a continuous strictly monotone function is continuous and strictly monotone. Thus $P(\theta)$ is continuous on $(-\pi/2, \pi/2)$. Using the *doubling property* (4.2), $P(\theta)$ is continuous on all of $(-\pi, \pi)$.

Since $\theta(J) = \pi/2$, $P(\pi/2) = J$. By (2.3), $P(\theta)$ satisfies (4.1) on $(-\pi/2, \pi/2)$, hence (4.2) holds for θ in $(-\pi, \pi)$.

Fix $\alpha \ge \pi$, and suppose $P(\theta)$ is defined on $(-\alpha, \alpha)$ and satisfies (4.1) on $(-\alpha/2, \alpha/2)$. If $\hat{P}(\theta)$ extends $P(\theta)$ to $(-2\alpha, 2\alpha)$ and satisfies (4.1) on $(-\alpha, \alpha)$, then $\hat{P}(\theta) = \hat{P}(\theta/2)^2 = P(\theta/2)^2$ on $(-2\alpha, 2\alpha)$, hence $\hat{P}(\theta)$ is uniquely determined. Conversely, define $\hat{P}(\theta) = P(\theta/2)^2$ on $(-2\alpha, 2\alpha)$. Then $\hat{P}(\theta)$ satisfies (4.1) on $(-\alpha, \alpha)$, since

$$\begin{aligned} \hat{P}(\theta)\hat{P}(\theta') &= P(\theta/2)^2 P(\theta'/2)^2 \\ &= P((\theta+\theta')/2)^2 = \hat{P}(\theta+\theta'), \end{aligned}$$

for θ, θ' in $(-\alpha, \alpha)$, and $\hat{P}(\theta)$ extends $P(\theta)$, since $\hat{P}(\theta) = P(\theta/2)^2 = P(\theta)$ on $(-\alpha, \alpha)$.

Iterating this, $P(\theta)$ can be extended uniquely to $(-2\pi, 2\pi)$, $(-4\pi, 4\pi)$, $(-8\pi, 8\pi)$, ..., with the extensions satisfying (4.1) on $(-\pi, \pi)$, $(-2\pi, 2\pi)$, $(-4\pi, 4\pi)$, ..., resulting in a continuous map $P(\theta)$ satisfying (4.1) for all reals. Since $P(\pi) = P(\pi/2)^2 = J^2 = -N$, $P(\theta)$ is surjective. Since $\theta(P)$ is bijective and $P(2\pi) = P(\pi)^2 = N$, 2π is the least positive period of $P(\theta)$. We write $P(\theta) = (\cos\theta, \sin\theta)$ for the extension.

Proof of uniqueness of $P(\theta)$ satisfying (4.1), step 1. Let $P(\theta)$ be any non-constant continuous map $P(\theta)$ of the real line into the unit circle satisfying (4.1). *Then $P(\theta)$ has a positive period:*

Write $P(\theta) = (x(\theta), y(\theta))$. Since 0 is a period, $x(0) = 1$. Since $P(\theta)$ is non-constant, there is an $\alpha \neq 0$ with $x(\alpha) < 1$. Since $\cos 0 = 1$, by continuity, there is an $n \ge 1$ with $x(\alpha) < \cos(2\pi/n) < 1$. By the intermediate value theorem, there is a $\beta \neq 0$ with $x(\beta) = \cos(2\pi/n)$, hence $y(\beta)$ equals one of $\pm\sin(2\pi/n)$. By (4.1),

$$\begin{aligned} P(n\beta) = P(\beta)^n &= (\cos(2\pi/n), \pm\sin(2\pi/n))^n \\ &= (\cos(2\pi), \pm\sin(2\pi)) = N, \end{aligned}$$

hence $P(\pm n\beta) = N$, hence there is a positive period.

[2] A more conventional definition of *period* requires $P(\theta+\omega) = P(\theta)$, for every θ. Because of the addition formula (4.1), $P(\theta+\omega) = P(\theta)P(\omega)$, so the two definitions are equivalent in our context.

Proof of uniqueness of $P(\theta)$ satisfying (4.1)*, step 2. $P(\theta)$ has a least positive period ω.*

Since a limit of periods is a period, the infimum ω of all positive periods is a period. For every real θ and every period $\beta > 0$, there is some integer n for which $n\beta \leq \theta < (n+1)\beta$. Hence for every period $\beta > 0$, every real θ lies within distance β of some period. If $\omega = 0$, there would be arbitrarily small positive periods β. This would imply every real θ is the limit of some sequence of periods, and thus $P(\theta) \equiv N$. Since identically constant functions are disallowed, $\omega > 0$.

Proof of uniqueness of $P(\theta)$ satisfying (4.1)*, step 3.* By rescaling $P(\theta)$ to $P(\omega\theta/2\pi)$, we may assume $\omega = 2\pi$. Then $P(\pi) = -N$, so $P(\pi/2) = J$ or $P(-\pi/2) = J$. By rescaling $P(\theta)$ to $P(-\theta)$ if necessary, we may assume $P(\pi/2) = J$. *Then $P(\theta)$ is uniquely determined:*

Since 2π is the least positive period, $P(\theta) \neq \pm N$ for $0 < |\theta| < \pi$. In particular, $\sqrt{P(\theta)}$ is defined for $-\pi < \theta < \pi$. Since (4.1) implies (4.2),

$$P(\theta/2) = \sqrt{P(\theta)}, \qquad -\pi < \theta < \pi, \tag{4.4}$$

up to sign. We claim (4.4) is correct as written, with no sign adjustment. This is immediate when $\theta = 0$, so assume $\theta \neq 0$. If the imaginary parts of $P(\theta)$ and $P(\theta/2)$ have opposite signs, then, by the intermediate value theorem, for some θ' between θ and $\theta/2$, we have $P(\theta') = N$ or $P(\theta') = -N$. Since this can't happen, the imaginary parts of $P(\theta)$ and $P(\theta/2)$ must have the same sign. It follows that the imaginary parts of $\sqrt{P(\theta)}$ and $P(\theta/2)$ have the same sign, establishing the claim. Using (4.1) and (4.4) repeatedly, $P(\theta)$ satisfies (4.3) for all dyadic rationals $\theta = k/2^n$. Since the dyadic rationals are dense and $P(\theta)$ is continuous, this determines $P(\theta)$. □

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